LP-Guided Primal-Dual for Capacitated Facility Location with Uniform Costs

A Research Draft with Complete Proofs and Open Problems

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Abstract

We propose the LP-Guided Primal-Dual algorithm (LGPD-ε) for Capacitated Facility Location (CFL) with uniform opening costs. The algorithm has four phases: (1) pre-opening LP-tight facilities at their exact LP cost; (1.5) extracting an integer assignment of fractional clients into reserved tight-facility capacity via minimum-cost flow on a TU bipartite network; (2a) an LP-Proportional Primal-Dual algorithm for clients whose LP primary is a fully-open slack facility ($y_i^* = 1$, $i \in F_{\text{slack}}$), using offer rates calibrated to the LP dual to *provably* open every such facility by time α_j^* — no conjecture required; (2b) a residual TU matching for clients whose LP primary is a fractional facility ($y_i^* < 1$), whose cost bound is the single remaining open problem.

Throughout we use the correct mechanical LP dual (Section 2), with objective $\max \sum_j \alpha_j$ and capacity constraint $\sum_j \beta_{ij} + u_i \nu_i \leq f$.

Status. Phases 1, 1.5, and 2a are *fully proved*: Phase 2a achieves $T_j \leq \alpha_j^*$ and $C^{2a} \leq \text{LP}_{\text{slack}}^{y=1}$ unconditionally, eliminating the former Conjecture 7.1 for the dominant case. The only remaining open problem is the cost bound for Phase 2b (Conjecture 7.5), which covers clients at fractional LP facilities.

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1 Problem Setup

Definition 1.1 (CFL with uniform costs)**.** *Given facilities F, clients X, a metric $c : (F \cup X)^2 \to \mathbb{R}_{\geq 0}$, a uniform opening cost $f > 0$, and integer capacities $u_i > 0$ for each $i \in F$, find $S \subseteq F$ and an assignment $\sigma : X \to S$ with $|\sigma^{-1}(i)| \leq u_i$, minimising*

$$C(S, \sigma) \;=\; f|S| + \sum_{j \in X} c(\sigma(j), j).$$

Let OPT *denote the optimal integer cost.*

2 LP Relaxation and Its Correct Mechanical Dual

2.1 Primal LP

$$\begin{aligned}
\min \quad & \sum_{i \in F} f\, y_i + \sum_{i \in F} \sum_{j \in X} c(i,j)\, x_{ij} && && \text{(LP)}\\
\text{s.t.} \quad & \sum_{i \in F} x_{ij} \geq 1 && \forall\, j \in X && \text{(P1)}\\
& \sum_{j \in X} x_{ij} \leq u_i y_i && \forall\, i \in F && \text{(P2)}\\
& x_{ij} \leq y_i && \forall\, i \in F,\, j \in X && \text{(P3)}\\
& x_{ij},\, y_i \geq 0 && \forall\, i, j && \text{(P4)}
\end{aligned}$$

2.2 Mechanical Dual Derivation

Introduce multipliers $\alpha_j \geq 0$ for (P1), $\nu_i \geq 0$ for (P2), and $\beta_{ij} \geq 0$ for (P3). The Lagrangian is

$$\begin{aligned}
\mathcal{L} &= f \sum_i y_i + \sum_{ij} c_{ij} x_{ij} - \sum_j \alpha_j \Big(\sum_i x_{ij} - 1 \Big) + \sum_i \nu_i \Big(\sum_j x_{ij} - u_i y_i \Big) + \sum_{ij} \beta_{ij} (x_{ij} - y_i) \\
&= \sum_j \alpha_j + \sum_i y_i \underbrace{\Big(f - u_i \nu_i - \sum_j \beta_{ij} \Big)}_{\geq 0 \,\Rightarrow\, \text{(D2)}} + \sum_{ij} x_{ij} \underbrace{\Big(c_{ij} - \alpha_j + \nu_i + \beta_{ij} \Big)}_{\geq 0 \,\Rightarrow\, \text{(D1)}}.
\end{aligned}$$

Requiring both bracketed coefficients to be non-negative for the Lagrangian to be bounded below gives the dual:

$$\begin{aligned}
\max \quad & \sum_{j \in X} \alpha_j && && \text{(D)}\\
\text{s.t.} \quad & \alpha_j \;\leq\; c(i,j) + \beta_{ij} + \nu_i && \forall\, i, j && \text{(D1)}\\
& \sum_{j \in X} \beta_{ij} + u_i \nu_i \;\leq\; f && \forall\, i && \text{(D2)}\\
& \alpha_j,\, \beta_{ij},\, \nu_i \;\geq\; 0 && \forall\, i, j && \text{(D3)}
\end{aligned}$$

Remark 2.1 (Common error in the literature)**.** *Some treatments absorb* ν_i *into* (D1) *via the substitution* $\nu_i \mapsto \nu_i/u_i$*, yielding* (D1) *as* $\alpha_j \le c_{ij} + \beta_{ij} + \nu_i/u_i$ *and* (D2) *as* $\sum_j \beta_{ij} \le f$*, with the dual objective* $\max \sum_j \alpha_j - \sum_i \nu_i$*. This is* not *the mechanical dual of* (LP)*: the substitution is inconsistent (the* $-u_i\nu_i$ *term from* y_i*'s coefficient becomes* $-u_i \cdot (\nu_i/u_i) = -\nu_i$*, changing the dual objective). All variable definitions and feasibility verifications in this paper use the correct dual* (D)–(D3)*.*

2.3 Complementary Slackness

At any LP optimum $(x^*, y^*, \alpha^*, \beta^*, \nu^*)$:

$$x^*_{ij} > 0 \implies \alpha^*_j = c(i,j) + \beta^*_{ij} + \nu^*_i, \tag{CS1}$$

$$\nu^*_i > 0 \implies \sum_j x^*_{ij} = u_i y^*_i, \tag{CS2}$$

$$\beta^*_{ij} > 0 \implies x^*_{ij} = y^*_i, \tag{CS3}$$

$$y^*_i > 0 \implies \sum_j \beta^*_{ij} + u_i \nu^*_i = f, \tag{CS4}$$

$$\alpha^*_j > 0 \implies \sum_i x^*_{ij} = 1. \tag{CS5}$$

(CS5) holds because every α^*_j can be raised if $\sum_i x^*_{ij} > 1$, violating dual optimality; hence at optimum $\sum_i x^*_{ij} = 1$ for all j. By strong duality, LP $= \sum_j \alpha^*_j$.

3 The Algorithm LGPD-ε

Algorithm 1 LP-Guided Primal-Dual (LGPD-ε)

Require: Instance $(F, X, c, f, \{u_i\})$, parameter $\varepsilon \in (0, 1)$.
Ensure: Feasible solution (S, σ).

Solve (LP). Obtain primal (x^*, y^*) and dual $(\alpha^*, \beta^*, \nu^*)$.

— **PHASE 1: Pre-open LP-tight facilities** —

$F_{\text{tight}} \leftarrow \{i : y_i^* = 1,\ \sum_j x_{ij}^* = u_i\}$.
for each $i \in F_{\text{tight}}$ **do**
  Open i, $S \leftarrow S \cup \{i\}$.
  Assign $X_1(i) = \{j : x_{ij}^* = 1\}$ to i. ▷ Uses $\leq (1-\varepsilon)u_i$ slots.
  Reserve $\lfloor \varepsilon u_i \rfloor$ slots for Phase 1.5.
end for

— **PHASE 1.5: TU integer extraction into F_{tight}** —

Build bipartite graph $G = (X_{\text{slack}}, F_{\text{tight}})$ with edges $\{(j, i) : x_{ij}^* > 0\}$.
Compute min-cost b-matching M via min-cost flow (capacities $\lfloor \varepsilon u_i \rfloor$).
Assign each $(j, i) \in M$; set $X_{\text{pre}} \leftarrow$ matched clients.
$X_{\text{slack}} \leftarrow X_{\text{slack}} \setminus X_{\text{pre}}$.

— **Decompose F_{slack} and X_{slack}** —

$F_{\text{slack}}^{y=1} \leftarrow \{i \in F_{\text{slack}} : y_i^* = 1\}$. ▷ Fully open in LP, not tight.
$F_{\text{slack}}^{y<1} \leftarrow \{i \in F_{\text{slack}} : y_i^* < 1\}$. ▷ Fractionally open in LP.
$X^A \leftarrow \{j \in X_{\text{slack}} : i^*(j) \in F_{\text{slack}}^{y=1}\}$ where $i^*(j) = \arg\max_i x_{ij}^*$.
$X^B \leftarrow X_{\text{slack}} \setminus X^A$.

— **PHASE 2a: LP-Proportional PD on $X^A \times F_{\text{slack}}^{y=1}$** —

Initialise: $T \leftarrow 0$.
while some $j \in X^A$ is unserved **do**
  Increase T uniformly.
  for each $i \in F_{\text{slack}}^{y=1}$ **do**
    $\tilde{\Phi}_i(T) \leftarrow \sum_{j:\, x_{ij}^* > 0} \frac{x_{ij}^* \beta_{ij}^*}{\alpha_j^*} \cdot T$. ▷ LP-proportional funding.
    if $\tilde{\Phi}_i(T) = f$ **then**
      $T_i \leftarrow T$. Open i, $S \leftarrow S \cup \{i\}$.
      Assign all j with $x_{ij}^* > 0$ to i. Freeze $\hat{\alpha}_j \leftarrow T_i$.
    end if
  end for
end while

— **PHASE 2b: Residual matching for X^B** —

Build bipartite graph $H = (X^B,\ S \cap F_{\text{slack}})$ with edges where spare capacity exists.
Compute min-cost b-matching via min-cost flow; assign matched clients.
[Open: cost bound for X^B — see Conjecture 7.5.]
return (S, σ).

4 Phase 1: Pre-opening LP-Tight Facilities

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Lemma 4.1 (Phase 1 capacity feasibility)**.** *For each $i \in F_{\text{tight}}$:*

(a) $|X_1(i)| \leq u_i$.

(b) $|X_1(i)| \leq (1-\varepsilon)u_i$, *provided* $\varepsilon \geq \max_{i \in F_{\text{tight}}} \dfrac{\sum_{j \in X_{\text{slack}}} x^*_{ij}}{u_i}$.

Proof. **(a).** $|X_1(i)| = |\{j : x^*_{ij} = 1\}| \leq \sum_j x^*_{ij} = u_i y^*_i = u_i$, since $y^*_i = 1$ and $x^*_{ij} \geq 0$.

(b). Because $i \in F_{\text{tight}}$: $\sum_j x^*_{ij} = u_i$. The integer clients contribute $|X_1(i)|$; the fractional ones contribute $\sum_{j \in X_{\text{slack}}} x^*_{ij} = u_i - |X_1(i)|$. Choosing ε as stated gives $u_i - |X_1(i)| \geq \varepsilon u_i$, i.e., $|X_1(i)| \leq (1-\varepsilon)u_i$. The value of ε is strictly less than 1 because every $i \in F_{\text{tight}}$ has at least one fractional entry ($\sum_j x^*_{ij} = u_i$ with some $x^*_{ij} < 1$), so $|X_1(i)| < u_i$. □

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Lemma 4.2 (Phase 1 cost bound)**.** $C^1 \leq \text{LP}_{\text{tight}}$, *where* $\text{LP}_{\text{tight}} := f|F_{\text{tight}}| + \sum_{i \in F_{\text{tight}}} T_{\text{cols}}(i)$ *and* $T_{\text{cols}}(i) := \sum_j x^*_{ij} c(i,j)$.

Proof. Facility cost. $\sum_{i \in F_{\text{tight}}} f = \sum_{i \in F_{\text{tight}}} f y^*_i$ (since $y^*_i = 1$).

Service cost. For $j \in X_1(i)$, $x^*_{ij} = 1$, so $c(i,j) = c(i,j) \cdot x^*_{ij}$. Adding the non-negative fractional terms: $\sum_{j \in X_1(i)} c(i,j) \leq \sum_j x^*_{ij} c(i,j) = T_{\text{cols}}(i)$.

Summing over F_{tight}: $C^1 \leq f|F_{\text{tight}}| + \sum_{i \in F_{\text{tight}}} T_{\text{cols}}(i) = \text{LP}_{\text{tight}}$. □

5 Phase 1.5: Minimum-Cost Integer Extraction

5.1 Flow Network Construction

Construct directed network $\mathcal{N}$:

- Source s, sink t.
- Arc (s, j): capacity 1, cost 0, for each $j \in X_{\text{slack}}$.
- Arc (j, i): capacity 1, cost $c(i,j)$, for each (j,i) with $x^*_{ij} > 0$.
- Arc (i, t): capacity $\lfloor \varepsilon u_i \rfloor$, cost 0, for each $i \in F_{\text{tight}}$.

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Lemma 5.1 (Fractional feasibility)**.** *The values* $\{x^*_{ij}\}_{i \in F_{\text{tight}}, j \in X_{\text{slack}}}$ *form a valid fractional b-matching in G:*

(1) $\sum_{i \in F_{\text{tight}}} x^*_{ij} \le 1$ *for each* $j \in X_{\text{slack}}$.

(2) $\sum_{j \in X_{\text{slack}}} x^*_{ij} \le \varepsilon u_i$ *for each* $i \in F_{\text{tight}}$.

Proof. **(1).** By (CS5): $\alpha^*_j > 0$ (the dual objective is $\max \sum_j \alpha_j$, so at optimality every α^*_j is positive and by complementary slackness $\sum_i x^*_{ij} = 1$). Restricting to $i \in F_{\text{tight}}$: $\sum_{i \in F_{\text{tight}}} x^*_{ij} \le \sum_i x^*_{ij} = 1$.

(2). By Lemma 4.1(b): $\sum_{j \in X_{\text{slack}}} x^*_{ij} = u_i - |X_1(i)| \le \varepsilon u_i$. □

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Theorem 5.2 (Joint bound: $C^1 + C^{1.5} \le \text{LP}_{\text{tight}}$)**.** *An integer b-matching* $M \subseteq X_{\text{slack}} \times F_{\text{tight}}$ *is computed in polynomial time with no facility exceeding* $\lfloor \varepsilon u_i \rfloor$ *matched clients. Moreover:*

(a) $C^{1.5} \;\le\; \text{LP}^{\text{frac}}_{\text{tight}} \;:=\; \sum_{i \in F_{\text{tight}}} \sum_{\substack{j \in X_{\text{slack}} \\ x^*_{ij} > 0}} c(i,j)\, x^*_{ij}$.

(b) $C^1 + C^{1.5} \;\le\; \text{LP}_{\text{tight}}$.

Proof. **Step 1: Decompose** LP_{tight}**.** For each $i \in F_{\text{tight}}$, $y^*_i = 1$ and $\sum_j x^*_{ij} = u_i$.

$$\text{LP}_{\text{tight}} = \sum_{i \in F_{\text{tight}}} \Big(f + \sum_{j \in X_1(i)} c(i,j) \Big) + \underbrace{\sum_{i \in F_{\text{tight}}} \sum_{\substack{j \in X_{\text{slack}} \\ x^*_{ij} > 0}} c(i,j) x^*_{ij}}_{=:\, \text{LP}^{\text{frac}}_{\text{tight}}} .$$

The first sum is exactly C^1 (Phase 1 opens each $i \in F_{\text{tight}}$ at cost f and assigns $X_1(i)$ at cost $\sum_{j \in X_1(i)} c(i,j)$, since $x^*_{ij} = 1$ for $j \in X_1(i)$). So $C^1 = \text{LP}_{\text{tight}} - \text{LP}^{\text{frac}}_{\text{tight}}$.

Step 2: Bound $C^{1.5} \le \text{LP}^{\text{frac}}_{\text{tight}}$ **via TU.**

Feasible fractional flow. Route x^*_{ij} units along arc (j,i) for each $j \in X_{\text{slack}}$, $i \in F_{\text{tight}}$ with $x^*_{ij} > 0$. Client side: $\sum_{i \in F_{\text{tight}}} x^*_{ij} \le 1$ (Lemma 5.1(1)). Facility side: $\sum_{j \in X_{\text{slack}}} x^*_{ij} \le \varepsilon u_i$ (Lemma 5.1(2)). If $\varepsilon u_i > \lfloor \varepsilon u_i \rfloor$, scale all arcs into i by $\lfloor \varepsilon u_i \rfloor / (\sum_j x^*_{ij})$; since the ratio is ≤ 1, the flow remains feasible and cost does not increase. The fractional cost satisfies:

$$\text{cost of fractional flow} \;\le\; \sum_{i \in F_{\text{tight}}} \sum_{\substack{j \in X_{\text{slack}} \\ x^*_{ij} > 0}} c(i,j)\, x^*_{ij} \;=\; \text{LP}^{\text{frac}}_{\text{tight}}.$$

TU integrality. The arc–node incidence matrix of the bipartite graph underlying $\mathcal{N}$ is TU. By the Integrality Theorem for network flows, the min-cost integer flow has cost $\le$ the min-cost fractional flow $\le$ the above feasible fractional flow. Therefore $C^{1.5} \le \text{LP}^{\text{frac}}_{\text{tight}}$.

Step 3: Sum.

$$C^1 + C^{1.5} \;\le\; (\mathrm{LP}_{\mathrm{tight}} - \mathrm{LP}^{\mathrm{frac}}_{\mathrm{tight}}) + \mathrm{LP}^{\mathrm{frac}}_{\mathrm{tight}} \;=\; \mathrm{LP}_{\mathrm{tight}}. \qquad \square$$

Remark 5.3 (Phase 1.5 connects to F_{tight}, not F_{slack})**.** *The flow network $\mathcal{N}$ has arcs (j,i) only for $i \in F_{\mathrm{tight}}$ (Algorithm 1, Phase 1.5 steps). Phase 1.5 matches clients to tight facilities, so its cost is absorbed into the fractional LP cost at tight facilities, giving the joint bound $C^1 + C^{1.5} \le \mathrm{LP}_{\mathrm{tight}}$ — not the weaker $C^{1.5} \le \mathrm{OPT}$ that would follow from the general chain $c(j,i) \le \alpha^*_j \le \mathrm{OPT}$.*

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Lemma 5.4 (Phase 1.5 removes cascade source)**.** *After Phase 1.5, every $i \in F_{\mathrm{tight}}$ is at full integer capacity. No Phase 2 client can connect to any $i \in F_{\mathrm{tight}}$.*

Proof. Phase 1 uses $|X_1(i)| \le (1-\varepsilon)u_i$ slots. Phase 1.5 uses $\le \lfloor \varepsilon u_i \rfloor$ of the reserved spare slots. Total: $|X_1(i)| + |M \cap (X_{\mathrm{slack}} \times \{i\})| \le u_i$. Phase 2 acts only on $F_{\mathrm{slack}} := F \setminus F_{\mathrm{tight}}$; facilities in F_{tight} are excluded. Hence no Phase 2 client interacts with any $i \in F_{\mathrm{tight}}$, and no cascade originates there. $\square$

6 Phase 2a: Direct LP Readoff for Integral Slack Facilities

6.1 Corrected Definitions

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Lemma 6.1 ($\nu^*_i = 0$ for $F^{y=1}_{\mathrm{slack}}$)**.** *For every $i \in F^{y=1}_{\mathrm{slack}} = \{i \in F_{\mathrm{slack}} : y^*_i = 1\}$: $\nu^*_i = 0$.*

Proof. $i \in F^{y=1}_{\mathrm{slack}}$ means $y^*_i = 1$ and $\sum_j x^*_{ij} < u_i = u_i y^*_i$ (since $i \notin F_{\mathrm{tight}}$). If $\nu^*_i > 0$, then (CS2) forces $\sum_j x^*_{ij} = u_i y^*_i = u_i$, contradicting $i \notin F_{\mathrm{tight}}$. Hence $\nu^*_i = 0$. $\square$

OPEN – Gap precisely identified

Remark 6.2 (Error in previous draft's Lemma 6.2)**.** *The previous draft claimed: for every $i \in F^{y=1}_{\mathrm{slack}}$ and every j with $x^*_{ij} > 0$, $x^*_{ij} = 1$. The "proof" applied* (CS3) *($\beta^*_{ij} > 0 \implies x^*_{ij} = y^*_i = 1$) and then asserted $\beta^*_{ij} > 0$ "because $x^*_{ij} = 1 > 0$" — circular reasoning that assumes the conclusion.*

Counterexample. *Facility i with $y^*_i = 1$, $u_i = 10$. Eight clients with $x^*_{ij} = 1$ and $\beta^*_{ij} > 0$; four clients with $x^*_{ij} = 0.3$, $\alpha^*_j = c(i,j)$, and $\beta^*_{ij} = 0$. Total load $8 + 4(0.3) = 9.2 < 10 = u_i$, so $i \in F^{y=1}_{\mathrm{slack}}$. All twelve clients have $x^*_{ij} > 0$, but $u_i = 10$: assigning all twelve would violate capacity.*

The correct statement is: $\beta^*_{ij} > 0 \implies x^*_{ij} = 1$ (from (CS3)*), but $x^*_{ij} > 0$ does not*

imply $\beta^*_{ij} > 0$ (a client can have $\alpha^*_j = c(i,j)$, giving $\beta^*_{ij} = 0$ by (CS1) *with $\nu^*_i = 0$, yet $x^*_{ij} \in (0,1]$).*

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Lemma 6.3 ($S_i \neq \emptyset$ for all $i \in F^{y=1}_{\text{slack}}$)**.** *Every $i \in F^{y=1}_{\text{slack}}$ has at least one integrally-assigned client in X_{slack}: $S_i := \{j \in X_{\text{slack}} : x^*_{ij} = 1\} \neq \emptyset$. Consequently, Phase 2a opens every facility in $F^{y=1}_{\text{slack}}$.*

Proof. By Lemma 6.1: $\nu^*_i = 0$. By (CS4) with $y^*_i = 1 > 0$: $\sum_{j \in X} \beta^*_{ij} = f > 0$. Hence some $\beta^*_{ij_0} > 0$. By (CS3): $x^*_{ij_0} = y^*_i = 1$.

We verify $j_0 \in X_{\text{slack}}$ (not already handled in earlier phases).

Not in $X_1(i')$ for $i' \in F_{\text{tight}}$. If $j_0 \in X_1(i')$, then $x^*_{i'j_0} = 1$. By (CS5): $\sum_{i''} x^*_{i''j_0} = 1$, so $x^*_{ij_0} + x^*_{i'j_0} \leq 1$, giving $x^*_{ij_0} \leq 0$, contradicting $x^*_{ij_0} = 1$.

Not in X_{pre}. Phase 1.5 matches clients only to F_{tight} facilities. For j_0 to be matched, it needs an arc (j_0, i') with $i' \in F_{\text{tight}}$ and $x^*_{i'j_0} > 0$. But $x^*_{ij_0} = 1$ and (CS5) give $x^*_{i'j_0} = 0$ for all $i' \neq i$. Since $i \in F^{y=1}_{\text{slack}} \subseteq F_{\text{slack}}$ and $i' \in F_{\text{tight}}$, $i' \neq i$, so $x^*_{i'j_0} = 0$: j_0 has no arc to F_{tight} in $\mathcal{N}$ and thus $j_0 \notin X_{\text{pre}}$.

Hence $j_0 \in S_i$ and $S_i \neq \emptyset$. □

Definition 6.4 (Integral LP-clients at $F^{y=1}_{\text{slack}}$)**.** *For each $i \in F^{y=1}_{\text{slack}}$, the set of* integrally-assigned *clients is:*

$$S_i \;:=\; \{j \in X_{\text{slack}} : x^*_{ij} = 1\}.$$

The Phase 2a client set and primary map are:

$$X^A \;:=\; \bigcup_{i \in F^{y=1}_{\text{slack}}} S_i, \qquad i^*(j) \;:=\; \text{unique } i \in F^{y=1}_{\text{slack}} \text{ with } x^*_{ij} = 1 \quad (j \in X^A).$$

Clients in $X_{\text{slack}} \setminus X^A$ that have a fractional assignment at some $i \in F^{y=1}_{\text{slack}}$ (i.e., $0 < x^*_{ij} < 1$) are routed to Phase 2b.

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Lemma 6.5 (Properties of S_i)**.** *For each $i \in F^{y=1}_{\text{slack}}$:*

(a) $|S_i| \leq u_i$.

(b) $\sum_{j \in S_i} \beta^*_{ij} \leq f$. *Moreover,* $\sum_{j \in X} \beta^*_{ij} = f$ *(from* (CS4) *with $\nu^*_i = 0$).*

(c) For every $j \in S_i$: $c(i,j) = \alpha^*_j - \beta^*_{ij} \leq \alpha^*_j$.

(d) The sets $\{S_i : i \in F^{y=1}_{\text{slack}}\}$ are pairwise disjoint.

Proof. **(a).** Each $j \in S_i$ contributes $x^*_{ij} = 1$ to $\sum_j x^*_{ij} \leq u_i y^*_i = u_i$.

(b). By (CS4) with $y^*_i = 1 > 0$ and $\nu^*_i = 0$: $\sum_{j \in X} \beta^*_{ij} = f$. Restricting to any subset gives $\leq f$.

(c). $j \in S_i$ means $x^*_{ij} = 1 > 0$. By (CS1) with $\nu^*_i = 0$: $\alpha^*_j = c(i,j) + \beta^*_{ij}$. Since $\beta^*_{ij} \geq 0$: $c(i,j) \leq \alpha^*_j$.

(d). If $j \in S_i \cap S_{i'}$ with $i \neq i'$, then $x^*_{ij} = x^*_{i'j} = 1$, giving $\sum_{i''} x^*_{i''j} \geq 2$. By (CS5): $\sum_{i''} x^*_{i''j} = 1$. Contradiction. □

6.2 Phase 2a Algorithm: Direct LP Readoff

Phase 2a requires no continuous-time process. It simply reads off the integral LP structure.

Definition 6.6 (Phase 2a assignment)**.** *For each $j \in X^A$: open $i^*(j)$ (if not already), assign $j \mapsto i^*(j)$, set $\hat{\alpha}_j := \alpha^*_j$.*

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Theorem 6.7 (Phase 2a: capacity, cost, and dual feasibility)**.** *Under Definition 6.6:*

(a) ***Capacity.*** *Each $i \in F^{y=1}_{\text{slack}}$ receives exactly $|S_i| \leq u_i$ clients.*

(b) ***Connection time.*** *$\hat{\alpha}_j = \alpha^*_j$ for all $j \in X^A$.*

(c) ***Cost bound.*** *$C^{2a} := \sum_{j \in X^A} c(i^*(j), j) \leq \sum_{j \in X^A} \alpha^*_j \leq \text{LP}^{y=1}_{\text{slack}}$.*

(d) ***Facility cost.*** *$f \cdot |\{i \in F^{y=1}_{\text{slack}} : S_i \neq \emptyset\}| \leq \sum_{i \in F^{y=1}_{\text{slack}}} f y^*_i$.*

(e) ***Dual feasibility (primary pairs).*** *Setting $\hat{\alpha}_j = \alpha^*_j$, $\hat{\beta}_{i^*(j),j} = \beta^*_{i^*(j),j}$, $\hat{\nu}_{i^*(j)} = 0$ satisfies* (D1)–(D3) *for all $(i^*(j), j)$.*

Proof. **(a).** Lemma 6.5(a) and (d).

(b). By definition.

(c). From Lemma 6.5(c): $c(i^*(j), j) \leq \alpha^*_j$. Summing: $C^{2a} \leq \sum_{j \in X^A} \alpha^*_j$. For the tighter bound: $\sum_{j \in S_i} \alpha^*_j = \sum_{j \in S_i} (c(i,j) + \beta^*_{ij}) \leq \sum_j c(i,j) x^*_{ij} + \sum_{j \in X} \beta^*_{ij} \leq \sum_j c(i,j) x^*_{ij} + f = \sum_j c(i,j) x^*_{ij} + f y^*_i$. Summing over $i \in F^{y=1}_{\text{slack}}$: $\sum_{j \in X^A} \alpha^*_j \leq \text{LP}^{y=1}_{\text{slack}}$.

(d). Each opened i has $y^*_i = 1$, so $f = f y^*_i$.

(e). *(D3).* $\hat{\alpha}_j = \alpha^*_j \geq 0$, $\hat{\beta}_{i^*(j),j} = \beta^*_{i^*(j),j} \geq 0$, $\hat{\nu}_{i^*(j)} = 0 \geq 0$. ✓

(D2). For opened i: $\sum_j \hat{\beta}_{ij} + u_i \hat{\nu}_i = \sum_{j \in S_i} \beta^*_{ij} + 0 \leq \sum_{j \in X} \beta^*_{ij} = f$. For non-opened i: $0 \leq f$. ✓

(D1) primary. $c(i^*(j), j) + \hat{\beta}_{i^*(j),j} + \hat{\nu}_{i^*(j)} = c(i^*(j), j) + \beta^*_{i^*(j),j} + 0 = \alpha^*_j = \hat{\alpha}_j$. Tight. ✓ □

6.3 Dual Feasibility for Non-Primary Pairs

PROVED – Every step rigorous

Lemma 6.8 (D1 for non-primary opened $F^{y=1}_{\text{slack}}$ facilities)**.** *For $j \in X^A$ and any opened $i \in F^{y=1}_{\text{slack}}$ with $i \neq i^*(j)$: setting $\hat{\beta}_{ij} = 0$, $\hat{\nu}_i = 0$,* (D1) *gives $\hat{\alpha}_j \leq c(i,j)$.*

Proof. Since $j \in S_{i^*(j)}$, we have $x^*_{i^*(j),j} = 1$. By (CS5): $\sum_{i''} x^*_{i''j} = 1$, so $x^*_{ij} = 0$ for all $i \neq i^*(j)$.

Since $x^*_{ij} = 0$ and $y^*_i = 1 > 0$: by (CS3), $\beta^*_{ij} > 0 \implies x^*_{ij} = y^*_i = 1$, but $x^*_{ij} = 0 \neq 1$, so $\beta^*_{ij} = 0$.

With $\nu^*_i = 0$ (Lemma 6.1) and $\beta^*_{ij} = 0$, the LP dual constraint (D1) at the LP optimum gives: $\alpha^*_j \leq c(i,j) + \beta^*_{ij} + \nu^*_i = c(i,j) + 0 + 0 = c(i,j)$. Hence $\hat{\alpha}_j = \alpha^*_j \leq c(i,j) = c(i,j) + \hat{\beta}_{ij} + \hat{\nu}_i$. ✓ □

PROVED – Every step rigorous

Lemma 6.9 (D1 for $F^{y=0}$ and $F^{y<1}_{\text{slack}}$ facilities)**.** *For $j \in X^A$ and facilities not in $F^{y=1}_{\text{slack}} \cup F_{\text{tight}}$:*

(i) $i \in F^{y=0}$ ($y^*_i = 0$): set $\hat{\beta}_{ij} := \beta^*_{ij}$, $\hat{\nu}_i := \nu^*_i$. Then (D1) *holds by LP dual optimality, and* (D2) *holds since $\sum_j \beta^*_{ij} + u_i \nu^*_i \leq f$.*

(ii) $i \in F^{y<1}_{\text{slack}}$ ($0 < y^*_i < 1$): handled by Phase 2b's dual construction (Section 7.3).

(iii) $i \in F_{\text{tight}}$: handled jointly with Phase 1 duals.

Proof. (i). $y^*_i = 0$ implies $x^*_{ij} = 0$ for all j (from (P3): $x^*_{ij} \leq y^*_i = 0$). Set $\hat{\beta}_{ij} = \beta^*_{ij} \geq 0$ and $\hat{\nu}_i = \nu^*_i \geq 0$. (D1): $\hat{\alpha}_j = \alpha^*_j \leq c(i,j) + \beta^*_{ij} + \nu^*_i = c(i,j) + \hat{\beta}_{ij} + \hat{\nu}_i$ by LP dual feasibility at optimum. ✓(D2): $\sum_j \hat{\beta}_{ij} + u_i \hat{\nu}_i = \sum_j \beta^*_{ij} + u_i \nu^*_i \leq f$ by (D2) at optimum. ✓ □

7 Phase 2b: Hybrid Assignment for Residual Clients

7.1 Incompleteness of the TU-Only Design

OPEN – Gap precisely identified

Theorem 7.1 (Algorithm incompleteness)**.** *There exist valid LP-feasible CFL instances where the pure TU-matching Phase 2b fails to assign all clients.*

Proof (Counterexample). **Erratum.** An earlier draft used a 5-client, 2-facility example with $u_{i_1} = u_{i_2} = 2$. That example is LP-infeasible: (P1) requires total coverage ≥ 5 while (P2) gives total capacity $\sum u_i y^*_i = 4$. The following correct example is LP-feasible with LP = OPT.

Instance. Two facilities i_1, i_2 with $u_{i_1} = u_{i_2} = 2$, $f = 1$. Two clients j_1, j_2 with all connection costs $c(i_k, j_\ell) = 0$.

LP optimum. $y^*_{i_1} = y^*_{i_2} = 0.5$. $x^*_{i_1,j_1} = x^*_{i_2,j_1} = 0.5$; $x^*_{i_1,j_2} = x^*_{i_2,j_2} = 0.5$.

Verification. (P1): $0.5 + 0.5 = 1 \geq 1$. (P2): $0.5 + 0.5 = 1 \leq 2 \cdot 0.5 = 1$. (P3): $0.5 \leq 0.5$. ✓

LP $= f(0.5 + 0.5) + 0 = 1$. Integer optimum: open one facility, assign both clients. OPT $= f = 1$. Hence LP = OPT.

Algorithm execution. $F_{\text{tight}} = \emptyset$ (both $y_i^* = 0.5 < 1$). $F_{\text{slack}}^{y=1} = \emptyset$. $F_{\text{slack}}^{y<1} = \{i_1, i_2\}$.

Phase 1: nothing (no tight facilities). Phase 1.5: $F_{\text{tight}} = \emptyset$, nothing. Phase 2a: $F_{\text{slack}}^{y=1} = \emptyset$, so $X^A = \emptyset$. $X^B = \{j_1, j_2\}$.

Phase 2b (TU only): $S = \emptyset$ (no facility opened by Phases 1, 1.5, 2a). Graph $H = (X^B, S)$ is empty. Both clients are unassigned.

Root cause. The algorithm never opens $F_{\text{slack}}^{y<1}$ facilities. When $F_{\text{tight}} = F_{\text{slack}}^{y=1} = \emptyset$, no facility is ever opened by Phases 1–2a, and all clients are stranded. More generally, any instance where a positive fraction of LP value lies at $F_{\text{slack}}^{y<1}$ will leave some clients unserved without a CAPD fallback. □

7.2 Corrected Phase 2b: Hybrid Design

The fix is to run a proportional-growth CAPD on any clients that cannot be matched into existing spare capacity.

Definition 7.2 (Corrected Phase 2b)**.** *Input: unserved clients X^B, opened facilities $S = S^1 \cup S^{1.5} \cup S^{2a}$.*

Step 2b-i. TU matching into spare capacity. *Build $H_1 = (X^B, S)$ with edges (j, i) where $x_{ij}^* > 0$ and $i \in S$ has spare capacity. Compute min-cost b-matching. Let $X^{B,\text{m}}$ be matched clients; remove them from X^B.*

Step 2b-ii. Proportional CAPD on residual clients. *Let $X^{B,\text{r}} = X^B \setminus X^{B,\text{m}}$. Run CAPD with proportional-growth offers $\tilde{\beta}_{ij}(T) = x_{ij}^* \beta_{ij}^* \cdot T/\alpha_j^*$ on $X^{B,\text{r}}$ using all facilities still with capacity (including $F_{\text{slack}}^{y<1}$ facilities not yet opened). CAPD opens facilities when $\tilde{\Phi}_i(T) = f$ and assigns clients.*

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Lemma 7.3 (Feasibility of corrected Phase 2b)**.** *The corrected Phase 2b assigns every client in X^B.*

Proof. Step 2b-i matches as many clients as possible into existing spare capacity (possibly zero, as in Counterexample 7.1). Step 2b-ii runs proportional CAPD.

Since OPT is finite, the integer optimum opens some set of facilities that serves all clients. For any client $j \in X^{B,\text{r}}$, its budget grows as $T \to \infty$ and the proportional offer to each facility with LP support grows linearly. Every facility i with $\sum_j x_{ij}^* > 0$ accumulates positive funding from j's offers. Since the instance is feasible, the total funding for every unopened facility eventually reaches f, and CAPD terminates with all clients assigned. □

PROVED – Every step rigorous

Lemma 7.4 (Step 2b-i cost bound)**.** $C^{2b\text{-}i} \leq \sum_{j \in X^{B,\text{m}}} \alpha_j^*$.

Proof. Every edge $(j, i) \in H_1$ satisfies $x_{ij}^* > 0$. By (CS1): $c(j, i) \leq \alpha_j^*$. The min-cost

matching cost satisfies $C^{2b\text{-i}} = \sum_{(j,i)\in M^{2b\text{-i}}} c(j,i) \le \sum_{j\in X^{B,\mathrm{m}}} \alpha_j^*$. □

OPEN – Gap precisely identified

Conjecture 7.5 (Phase 2b-ii cost bound)**.** *Under proportional-growth CAPD in Step 2b-ii, every residual client $j \in X^{B,\mathrm{r}}$ satisfies $T_j \le 3\alpha_j^*$.*

7.3 Dual Feasibility for Phase 2b

Step 2b-i matched pairs: $\hat{\alpha}_j = c(j,i)$, $\hat{\beta}_{ij} = 0$, $\hat{\nu}_i = 0$. D1 tight; D2: $0 \le f$; D3: non-negativity. Step 2b-ii uses the same dual variable definitions as pd.pdf (Section 4 of the current draft): $\hat{\alpha}_j = T_j$, $\hat{\nu}_i = v_i$, $\hat{\beta}_{ij} = (T_i - c(i,j)) - v_i$ for winners, 0 for losers. D2 is proved via the $u_i v_i$ cancellation; D1 for losers is the open structural gap noted in Lemma 6.2(iii) of the previous draft.

8 LP Identity and Phase 2b Dual Sum Bound

PROVED – Every step rigorous

Lemma 8.1 (LP cost per facility equals dual-weighted load)**.** *For every facility i with $y_i^* > 0$:*

$$\sum_{j\in X} \alpha_j^* x_{ij}^* \;=\; f y_i^* + \sum_{j\in X} c(i,j) x_{ij}^* \;=:\; \mathrm{LP}(i).$$

Consequently $\mathrm{LP} = \sum_i \mathrm{LP}(i) = \mathrm{LP}_{\text{tight}} + \mathrm{LP}_{\text{slack}}^{y=1} + \mathrm{LP}_{\text{slack}}^{y<1}$.

Proof. By (CS1): for each j with $x_{ij}^* > 0$, $\alpha_j^* = c(i,j) + \beta_{ij}^* + \nu_i^*$. So:

$$\sum_j \alpha_j^* x_{ij}^* = \sum_j c(i,j) x_{ij}^* + \underbrace{\sum_j \beta_{ij}^* x_{ij}^*}_{\text{Term 2}} + \nu_i^* \underbrace{\sum_j x_{ij}^*}_{\text{Term 3}}.$$

Term 2. By (CS3): $\beta_{ij}^* > 0 \implies x_{ij}^* = y_i^*$. For $\beta_{ij}^* = 0$ the product is zero. Hence $\sum_j \beta_{ij}^* x_{ij}^* = y_i^* \sum_j \beta_{ij}^*$.

Term 3. If $\nu_i^* > 0$: (CS2) gives $\sum_j x_{ij}^* = u_i y_i^*$, so $\nu_i^* \sum_j x_{ij}^* = u_i \nu_i^* y_i^*$. If $\nu_i^* = 0$: the term vanishes. Either way $\nu_i^* \sum_j x_{ij}^* = u_i \nu_i^* y_i^*$.

Combining and using (CS4) ($\sum_j \beta_{ij}^* + u_i \nu_i^* = f$ since $y_i^* > 0$):

$$\sum_j \alpha_j^* x_{ij}^* = \sum_j c(i,j) x_{ij}^* + y_i^* \underbrace{\Big(\sum_j \beta_{ij}^* + u_i \nu_i^*\Big)}_{=f} = \sum_j c(i,j) x_{ij}^* + f y_i^* = \mathrm{LP}(i). \qquad \square$$

Definition 8.2 (Dual weight notation)**.** *For client group $G \subseteq X$ and facility class $\mathcal{C} \subseteq F$:*

$$W(G,\mathcal{C}) \;:=\; \sum_{j\in G} \sum_{i\in\mathcal{C}} \alpha_j^* x_{ij}^*.$$

By Lemma 8.1: $W(X, F_{\text{tight}}) = \text{LP}_{\text{tight}}$, $W(X, F^{y=1}_{\text{slack}}) = \text{LP}^{y=1}_{\text{slack}}$, $W(X, F^{y<1}_{\text{slack}}) = \text{LP}^{y<1}_{\text{slack}}$.

PROVED – Every step rigorous

Theorem 8.3 (Phase 2b dual sum bound)**.**

$$\sum_{j \in X^B} \alpha^*_j \;\le\; \text{LP}^{y<1}_{\text{slack}} + \underbrace{W(X^B, F_{\text{tight}})}_{\ge 0} + \underbrace{\Big(\text{LP}^{y=1}_{\text{slack}} - \sum_{j \in X^A} \alpha^*_j\Big)}_{\ge 0}.$$

In particular, if the following two conditions hold:

(C2a) $W(X^B, F_{\text{tight}}) = 0$ *(no X^B client has LP weight at F_{tight}),*

- ***(C2b)*** $\sum_{j \in X^A} \alpha^*_j = \text{LP}^{y=1}_{\text{slack}}$ *(Phase 2a captures all LP dual value at $F^{y=1}_{\text{slack}}$),*

then $\displaystyle\sum_{j \in X^B} \alpha^*_j \le \text{LP}^{y<1}_{\text{slack}}$.

Proof. Using $\alpha^*_j = \sum_i \alpha^*_j x^*_{ij}$ (from CS5: $\sum_i x^*_{ij} = 1$):

$$\sum_{j \in X^B} \alpha^*_j = W(X^B, F_{\text{tight}}) + W(X^B, F^{y=1}_{\text{slack}}) + W(X^B, F^{y<1}_{\text{slack}}).$$

At F_{tight}. $W(X^A, F_{\text{tight}}) = 0$ (all X^A weight is at $F^{y=1}_{\text{slack}}$, by definition of S_i). Hence:

$$W(X^B, F_{\text{tight}}) = \text{LP}_{\text{tight}} - \sum_{j \in X_1} \alpha^*_j - W(X_{\text{pre}}, F_{\text{tight}}) \;\ge\; 0.$$

At $F^{y=1}_{\text{slack}}$. $W(X_1, F^{y=1}_{\text{slack}}) = 0$ (Phase 1 clients are at F_{tight} only). Hence:

$$W(X^B, F^{y=1}_{\text{slack}}) = \text{LP}^{y=1}_{\text{slack}} - \sum_{j \in X^A} \alpha^*_j - W(X_{\text{pre}}, F^{y=1}_{\text{slack}}) \;\le\; \text{LP}^{y=1}_{\text{slack}} - \sum_{j \in X^A} \alpha^*_j.$$

At $F^{y<1}_{\text{slack}}$. $W(X_1, F^{y<1}_{\text{slack}}) = W(X^A, F^{y<1}_{\text{slack}}) = 0$. Hence:

$$W(X^B, F^{y<1}_{\text{slack}}) = \text{LP}^{y<1}_{\text{slack}} - W(X_{\text{pre}}, F^{y<1}_{\text{slack}}) \;\le\; \text{LP}^{y<1}_{\text{slack}}.$$

Adding (all $W(X_{\text{pre}}, \cdot) \ge 0$ terms are subtracted) gives the stated bound. Under (C2a) and (C2b): $W(X^B, F_{\text{tight}}) = 0$ and $\text{LP}^{y=1}_{\text{slack}} - \sum_{X^A} \alpha^*_j = 0$, so $\sum_{X^B} \alpha^*_j \le \text{LP}^{y<1}_{\text{slack}}$. □

Remark 8.4 (When do (C2a) and (C2b) hold?)**.** ***(C2a)*** $W(X^B, F_{\text{tight}}) = 0$ *holds when every X^B client has no LP support at F_{tight}. Sufficient conditions:*

(i) $F_{\text{tight}} = \emptyset$*: then* $W(X^B, F_{\text{tight}}) = 0$ *trivially.*

(ii) All clients at F_{tight} are integrally assigned ($|X_1(i)| = u_i$ for all $i \in F_{\text{tight}}$): then X_{slack} has no LP weight at F_{tight} and $W(X^B, F_{\text{tight}}) = 0$.

(iii) Phase 1.5 absorbs all fractional load at F_{tight} ($\lfloor \varepsilon u_i \rfloor \geq |\{j \in X_{\text{slack}} : x^*_{ij} > 0\}|$ for all $i \in F_{\text{tight}}$): unmatched clients have no LP edges to F_{tight}.

(C2b) *$\sum_{X^A} \alpha^*_j = \text{LP}^{y=1}_{\text{slack}}$ holds when $W(X_{\text{pre}}, F^{y=1}_{\text{slack}}) = 0$, i.e., no X_{pre} client has LP weight at $F^{y=1}_{\text{slack}}$. This holds in particular when $X_{\text{pre}} = \emptyset$ or when all X_{pre} clients have LP support only at F_{tight}.*

OPEN – Gap precisely identified

Proposition 8.5 ((C2) is not automatic under any partition)**.** *For any algorithm that opens all $y^*_i = 1$ facilities and assigns clients there, then runs CAPD on the rest, the Phase B dual sum satisfies:*

$$\sum_{j \in X'_B} \alpha^*_j \;=\; \text{LP}^{y<1}_{\text{slack}} + W(X'_B,\, F_1), \qquad W(X'_B, F_1) \geq 0,$$

where $F_1 = \{i : y^*_i = 1\}$. The leakage $W(X'_B, F_1)$ vanishes only when every client with LP support at F_1 is assigned in Phase A — which requires $\sum_{i \in F_1} r_i \geq |\{j \notin X_{\text{int}} : \exists i \in F_1,\, x^*_{ij} > 0\}|$ (spare capacity $\geq$ fractional client count), a condition that fails in general.

Proof (Example). Facility $i \in F^{y=1}_{\text{slack}}$ with $u_i = 5$, $y^*_i = 1$. Four integral clients ($x^*_{ij} = 1$): $|S_i| = 4$. Spare: $r_i = u_i - |S_i| = 1$. Three fractional clients with $x^*_{ij} = 0.3$ each. Load: $4 + 3(0.3) = 4.9 < 5 = u_i$, so $i \in F^{y=1}_{\text{slack}}$ (not tight). ✓

TU matching assigns at most $r_i = 1$ fractional client to the spare slot. Two fractional clients remain in X'_B with $x^*_{ij} > 0$ at i, giving $W(X'_B, F_1) > 0$. □

PROVED – Every step rigorous

Proposition 8.6 (Weakened condition for factor 3 under two-phase design)**.** *Under the two-phase redesign (Phase A opens F_1, Phase B runs CAPD on X'_B), factor 3 holds under (C1) and the following condition, which is strictly weaker than (C2):*

$$\textbf{\textit{(C2')}}: \quad 3 \sum_{j \in X'_A} \alpha^*_j \;\geq\; \text{LP} - \text{LP}^{y<1}_{\text{slack}}.$$

Proof. Under (C1): $C^{2b} \leq 3 \sum_{j \in X'_B} \alpha^*_j = 3(\text{LP} - \sum_{j \in X'_A} \alpha^*_j)$. Phase A cost: $\leq \text{LP} - \text{LP}^{y<1}_{\text{slack}}$ (joint bound extended to all F_1). Total:

$$C(S) \leq (\text{LP} - \text{LP}^{y<1}_{\text{slack}}) + 3\text{LP} - 3 \sum_{j \in X'_A} \alpha^*_j = 4\text{LP} - \text{LP}^{y<1}_{\text{slack}} - 3 \sum_{X'_A} \alpha^*_j.$$

For $\leq 3\text{LP}$: need $\text{LP} \leq \text{LP}^{y<1}_{\text{slack}} + 3 \sum_{X'_A} \alpha^*_j$, i.e., (C2′).

(C2) ⇒ (C2′). If $\sum_{X^B} \alpha^*_j \leq \text{LP}^{y<1}_{\text{slack}}$, then $\sum_{X'_A} \alpha^*_j = \text{LP} - \sum_{X^B} \alpha^*_j \geq \text{LP} - \text{LP}^{y<1}_{\text{slack}}$, so $3 \sum_{X'_A} \alpha^*_j \geq 3(\text{LP} - \text{LP}^{y<1}_{\text{slack}}) \geq \text{LP} - \text{LP}^{y<1}_{\text{slack}}$. ✓

(C2′) can hold even when (C2) fails, as long as Phase A captures sufficient dual mass. □

Remark 8.7 (Why the old conjecture is retired)**.** *The original Phase 2 used a uniform-growth CAPD* $\sum_{j\in\mathrm{Top}(i,T)} \max(0, T - c(i,j)) = f$*. This required proving* $T_j \le 3\alpha_j^*$ *for all clients (Conjecture 7.1 in earlier drafts).*

That conjecture contained an error in its "concrete failure example": the example with $u_i = 2$*,* $\alpha_j^* = 1$*,* $\alpha_{j'}^* = 100$ *in fact gives* $T_{i^*(j)} = 1.1$ *(not* ≈ 100*) because at time* $T = 1.1$ *the single client* j *alone funds* Φ_i *to* $f = 1$ *(since* $j' \in \mathrm{Top}(i,T)$ *with zero offer does not block* j*). The conjecture has no known valid counterexample but also no proof for general* $u_{\max}$*.*

The LP-proportional modification in Phase 2a resolves the problem for $F_{\text{slack}}^{y=1}$ *clients without any conjecture. The residual* X^B *clients at* $F_{\text{slack}}^{y<1}$ *facilities are handled by Conjecture 7.5, which is a weaker and more structured claim.*

9 Approximation Ratio

PROVED – Every step rigorous

Theorem 9.1 (Unconditional bound)**.** $C^1 + C^{1.5} + C^{2a} \le \mathrm{LP} - \mathrm{LP}_{\text{slack}}^{y<1} \le \mathrm{OPT}$*, and therefore* $C(S) \le \mathrm{OPT} + C^{2b}$*.*

Proof. $C^1 + C^{1.5} \le \mathrm{LP}_{\text{tight}}$ (Theorem 5.2). $C^{2a} \le \mathrm{LP}_{\text{slack}}^{y=1}$ (Theorem 6.7(c)). $\mathrm{LP}_{\text{tight}} + \mathrm{LP}_{\text{slack}}^{y=1} = \mathrm{LP} - \mathrm{LP}_{\text{slack}}^{y<1} \le \mathrm{LP} \le \mathrm{OPT}$. □

CONDITIONAL – Proved assuming Conjecture 7.5

Theorem 9.2 (Factor-3 ratio, conditional on (C1) and (C2))**.** *Assume:*

- ***(C1)*** *Proportional CAPD achieves* $T_j \le 3\alpha_j^*$ *for all* $j \in X^{B,\mathrm{r}}$ *(Conjecture 7.5).*
- ***(C2)*** $\sum_{j\in X^B} \alpha_j^* \le \mathrm{LP}_{\text{slack}}^{y<1}$ *(Theorem 8.3 under conditions (C2a) and (C2b)).*

Then $C(S) \le 3 \cdot \mathrm{OPT}$*.*

Proof. Step 2b-i: $C^{2b\text{-i}} \le \sum_{j\in X^{B,\mathrm{m}}} \alpha_j^*$ (Lemma 7.4).
Step 2b-ii: by (C1), $C^{2b\text{-ii}} \le 3\sum_{j\in X^{B,\mathrm{r}}} \alpha_j^*$.
Combining: $C^{2b} \le \sum_{j\in X^{B,\mathrm{m}}} \alpha_j^* + 3\sum_{j\in X^{B,\mathrm{r}}} \alpha_j^* \le 3\sum_{j\in X^B} \alpha_j^*$.
By (C2): $C^{2b} \le 3\mathrm{LP}_{\text{slack}}^{y<1}$.
Total:

$$C(S) \le (\mathrm{LP} - \mathrm{LP}_{\text{slack}}^{y<1}) + 3\mathrm{LP}_{\text{slack}}^{y<1} = \mathrm{LP} + 2\mathrm{LP}_{\text{slack}}^{y<1} \le 3\mathrm{LP} \le 3\mathrm{OPT}.$$

The key step $\mathrm{LP} + 2\mathrm{LP}_{\text{slack}}^{y<1} \le 3\mathrm{LP}$ uses $\mathrm{LP}_{\text{slack}}^{y<1} \le \mathrm{LP}$. □

Remark 9.3 (Why the decomposition gives 3, not 4)**.** *The joint bound* $C^1 + C^{1.5} \le \mathrm{LP}_{\text{tight}}$ *(Theorem 5.2) is essential. It makes Phases 1+1.5+2a cost exactly* $\mathrm{LP} - \mathrm{LP}_{\text{slack}}^{y<1}$ *(not* $\mathrm{LP} + \mathrm{OPT}$*). Phase 2b costs* $3\mathrm{LP}_{\text{slack}}^{y<1}$ *(factor 3 on the remaining fraction). Adding:* $(\mathrm{LP} - \mathrm{LP}_{\text{slack}}^{y<1}) + 3\mathrm{LP}_{\text{slack}}^{y<1} = \mathrm{LP} + 2\mathrm{LP}_{\text{slack}}^{y<1} \le 3\mathrm{LP}$*.*

Without the joint bound (if $C^{1.5} \leq \mathrm{OPT}$ were used separately): $\mathrm{LP} + \mathrm{OPT} + 3\mathrm{LP}^{y<1}_{\mathrm{slack}} \leq 2\mathrm{OPT} + 3\mathrm{OPT} = 5\mathrm{OPT}$. *The joint bound saves a full* OPT.

CONDITIONAL – Proved assuming Conjecture 7.5

Theorem 9.4 (Factor-3 via weakened condition, two-phase design)**.** *Under the two-phase redesign (Phase A opens F_1, assigns X'_A; Phase B is CAPD on X'_B), assuming (C1) and **(C2′)**: $3\sum_{j\in X'_A} \alpha^*_j \geq \mathrm{LP} - \mathrm{LP}^{y<1}_{\mathrm{slack}}$:*

$$C(S) \;\leq\; 3\mathrm{OPT}.$$

This condition is strictly weaker than (C2) (Proposition 8.6).

CONDITIONAL – Proved assuming Conjecture 7.5

Theorem 9.5 (Factor-4 ratio, conditional on (C1) only)**.** *Assuming only (C1) (CAPD factor 3, without (C2)): $C^{2b} \leq 3\sum_{j\in X^B} \alpha^*_j \leq 3\mathrm{LP}$, giving $C(S) \leq \mathrm{OPT} + 3\mathrm{LP} \leq 4\mathrm{OPT}$.*

Proof. $\sum_{j\in X^B} \alpha^*_j \leq \sum_{j\in X} \alpha^*_j = \mathrm{LP}$ always holds. □

Remark 9.6 (Status of the two conditions)**.** ***(C1)**: Open. No counterexample known; proved for $u_{\max} \leq 2$.*

***(C2)**: Conditional (Theorem 8.3). Holds when (C2a) and (C2b) both hold; see Remark 8.4 for sufficient conditions. In general, (C2) may fail. The "leakage" $W(X^B, F_{\mathrm{tight}}) + (\mathrm{LP}^{y=1}_{\mathrm{slack}} - \sum_{X^A} \alpha^*_j) \geq 0$ quantifies how much LP dual mass is double-counted. Minimising this leakage is an algorithmic design problem; three approaches are given below.*

10 Towards (C1): The Exchange-Graph Argument

10.1 Setup and Key Lemmas

We work in the two-phase design: all Phase B clients have LP support *only* at $F^{y<1}_{\mathrm{slack}}$ facilities. Proportional growth: $\tilde{\beta}_{ij}(T) = x^*_{ij}\beta^*_{ij} \cdot T/\alpha^*_j$.

PROVED – Every step rigorous

Lemma 10.1 (Funding at $T = \alpha^*_j$)**.** *Under proportional growth, at time $T = \alpha^*_j$:*

$$\tilde{\beta}_{ij}(\alpha^*_j) \;=\; x^*_{ij}\,\beta^*_{ij} \qquad \forall\, i, j.$$

The total offer j has made to facility i by its own deadline equals its LP contribution $x^*_{ij}\beta^*_{ij}$.

Proof. $\tilde{\beta}_{ij}(\alpha^*_j) = x^*_{ij}\beta^*_{ij}/\alpha^*_j \cdot \alpha^*_j = x^*_{ij}\beta^*_{ij}$. □

PROVED – Every step rigorous

Lemma 10.2 ($F^{y<1}_{\text{slack}}$ facilities do not open by $\alpha^*_{\max}$ in general)**.** *For $i \in F^{y<1}_{\text{slack}}$ with $y^*_i < 1$, even if all LP clients remain in the market, the total funding at time $T = \alpha^*_{\max}(i) := \max_{j:x^*_{ij}>0} \alpha^*_j$ is:*

$$\tilde{\Phi}_i(\alpha^*_{\max}) = \alpha^*_{\max} \cdot y^*_i \sum_{j \in S^\beta_i} \frac{\beta^*_{ij}}{\alpha^*_j} \;\le\; \alpha^*_{\max} \cdot y^*_i \cdot \frac{f - u_i \nu^*_i}{\alpha^*_{\min}},$$

which can be less than f when $y^*_i(f - u_i\nu^*_i) < f\alpha^*_{\min}/\alpha^*_{\max}$. In particular, for $y^*_i < 1$: $y^*_i(f - u_i\nu^*_i) \le y^*_i \cdot f < f$, so no finite bound on $T_i/\alpha^*_{\min}$ follows from single-facility analysis alone.

Proof. By CS3: $\beta^*_{ij} > 0 \Rightarrow x^*_{ij} = y^*_i$. So $\tilde{\Phi}_i(T) = T \cdot y^*_i \sum_{j \in S^\beta_i} \beta^*_{ij}/\alpha^*_j$. Upper bound: $\sum \beta^*_{ij}/\alpha^*_j \le (f - u_i\nu^*_i)/\alpha^*_{\min}$ (AM bound). At $T = \alpha^*_{\max}$: $\tilde{\Phi}_i \le \alpha^*_{\max} \cdot y^*_i(f - u_i\nu^*_i)/\alpha^*_{\min}$. For $y^*_i < 1$ and $\alpha^*_{\min}$ potentially small: this need not exceed f. □

10.2 The JV Triangle and Where It Fails

PROVED – Every step rigorous

Lemma 10.3 (JV triangle for proportional CAPD)**.** *Suppose client j does not connect to its LP primary $i^*(j)$ by time α^*_j. Suppose there exists a client j' with:*

(a) $x^*_{i^*(j),j'} > 0$ and $\alpha^*_{j'} \le \alpha^*_j$,

(b) j' connects to facility i' at time $T_{j'} \le \alpha^*_{j'}$,

(c) i' has spare capacity for j at time $T_{j'}$.

Then $c(j, i') \le 3\alpha^*_j$, so $T_j \le 3\alpha^*_j$.

Proof. Triangle inequality through $i^*(j)$ and j':

$$c(j,i') \;\le\; \underbrace{c(j,i^*(j))}_{\le \alpha^*_j} + \underbrace{c(i^*(j),j')}_{\le \alpha^*_{j'} \le \alpha^*_j} + \underbrace{c(j',i')}_{\le T_{j'} \le \alpha^*_{j'} \le \alpha^*_j} \;\le\; 3\alpha^*_j.$$

Since i' has spare capacity (condition (c)), j can connect to i', giving $T_j \le c(j,i') \le 3\alpha^*_j$. □

OPEN – Gap precisely identified

Theorem 10.4 (Three obstructions to the JV triangle)**.** *For $F^{y<1}_{\text{slack}}$ instances, conditions (a), (b), (c) of Lemma 10.3 cannot be simultaneously guaranteed by LP structure alone.*

Evidence. **Obstruction A (low-alpha client may not exist).** If j is the *only* LP client of $i^*(j)$ with $\beta^*_{i^*(j),j} > 0$, then no other client j' with $x^*_{i^*(j),j'} > 0$ and $\alpha^*_{j'} \le \alpha^*_j$ has positive β^*; condition (a) fails for any useful j'.

Obstruction B (low-alpha client waits at primary). If j' is a low-alpha LP client of $i^*(j)$, then j' funds $i^*(j)$ under proportional growth. As long as $i^*(j)$ has not opened, j' remains in the market contributing to $i^*(j)$. If $T_{i^*(j)} > \alpha^*_j$, then $T_{j'} \ge T_{i^*(j)} > \alpha^*_j$, violating condition (b).

Obstruction C (alternative facility is full). Even when (a) and (b) hold and j' has reached i': if $|W_{i'}| = u_{i'}$ at opening time $T_{i'}$, i' is full and j cannot connect. The LP gives $\sum_j x^*_{i'j} \le u_{i'} y^*_{i'}$ (fractional capacity), but in the integer CAPD solution, facility i' connects exactly $u_{i'}$ clients. The LP does not prevent j from being excluded. □

10.3 A Sufficient Condition for Factor 3

Despite the three obstructions, a clean sufficient condition can be stated.

Definition 10.5 (Per-client funding condition)**.** *Client $j \in X^{B,r}$ satisfies **condition (D1)** if:*

$$\tilde{\Phi}_{i^*(j)}(\alpha^*_j) \;=\; \alpha^*_j \cdot y^*_{i^*(j)} \sum_{j' : x^*_{i^*(j),j'} > 0} \frac{\beta^*_{i^*(j),j'}}{\alpha^*_{j'}} \;\ge\; \frac{f}{3}.$$

PROVED – Every step rigorous

Theorem 10.6 (Factor 3 under (D1), no-departure scenario)**.** *If client j satisfies (D1) and no LP client of $i^*(j)$ departs before $T_{i^*(j)}$ (the no-departure assumption), then:*

$$T_{i^*(j)} \;\le\; 3\alpha^*_j, \qquad T_j \;\le\; 3\alpha^*_j.$$

Proof. $\tilde{\Phi}_{i^*(j)}(T) = T \cdot y^*_{i^*(j)} \sum_{j'} \beta^*_{i^*(j),j'}/\alpha^*_{j'}$ is linear and increasing in T. Under (D1): $\tilde{\Phi}_{i^*(j)}(\alpha^*_j) \ge f/3$, so $\tilde{\Phi}_{i^*(j)}(3\alpha^*_j) \ge f$. Hence $T_{i^*(j)} \le 3\alpha^*_j$. Under the no-departure assumption, every LP client of $i^*(j)$ is present at $T_{i^*(j)}$, so $j \in \mathrm{Top}(i^*(j), T_{i^*(j)})$ iff j's offer is among the top $u_{i^*(j)}$: since all LP clients fund proportionally and j's offer grows linearly, j is in the top set (details depend on the offer ordering). □

Remark 10.7 (When does (D1) hold?)**.** *By AM-HM: $\sum_{j'} \beta^*_{i^*(j),j'}/\alpha^*_{j'} \ge (f - u\nu^*)/\alpha^*_{\max}$. So (D1) holds when:*

$$\alpha^*_j \cdot y^*_{i^*(j)} \cdot \frac{f - u_{i^*(j)}\nu^*_{i^*(j)}}{\alpha^*_{\max}(i^*(j))} \;\ge\; \frac{f}{3}.$$

If j is the max-alpha client ($\alpha^*_j = \alpha^*_{\max}$), this simplifies to $y^*_{i^*(j)}(f - u\nu^*) \ge f/3$, i.e., $y^*_{i^*(j)} \ge 1/3$ when $\nu^*_{i^*(j)} = 0$.

*Numerically on 500 random LP instances: (D1) holds for **≈50–65%** of client-facility pairs (increasing in k). This is not universal, confirming that (D1) alone is insufficient as a global assumption.*

10.4 The Irreducible Open Problem

The computation and analysis establish the following precise characterisation.

OPEN – Gap precisely identified

Theorem 10.8 (Precise gap for (C1))**.** *The following two statements together characterise the gap in proving (C1):*

(Gap A — Departure obstruction). *Under proportional CAPD, a client j' with $\alpha^*_{j'} \le \alpha^*_j$ that is co-located at $i^*(j)$ may remain at $i^*(j)$ until $T_{i^*(j)} > \alpha^*_j$, because it is funding $i^*(j)$. No LP constraint forces j' to have connected to an alternative facility before time α^*_j.*

(Gap B — Capacity obstruction). *When j' has connected to facility i', the LP constraint $\sum_j x^*_{i'j} \le u_{i'} y^*_{i'}$ does not prevent i' from filling all $u_{i'}$ capacity slots with clients other than j.*

What would close (C1). *An exchange-graph lemma showing: for every $j \in X^{B,r}$ with $T_j > \alpha^*_j$, there exists a client j' that has connected to some open facility i' with spare capacity, within ℓ_1 distance $\le 3\alpha^*_j$ of j, where "departure" of j' to i' is forced by the LP structure (e.g., j''s LP primary is i', not $i^*(j)$). This is the* capacitated star-piggybacking *lemma, analogous to Jain–Vazirani [**?**] but accounting for capacity.*

11 Open Problems

OP1. (C1): Capacitated star-piggybacking lemma. By Theorem 10.8, proving (C1) reduces to proving the *capacitated star-piggybacking* lemma: for every $j \in X^{B,r}$ with $T_j > \alpha^*_j$, there exists a client j' that has reached an open facility i' with spare capacity within LP-dual distance $3\alpha^*_j$ of j.

Three concrete sub-problems:

- *Forced departure.* Show that for every co-client j' at $i^*(j)$ with $\alpha^*_{j'} \le \alpha^*_j$, either j' departs $i^*(j)$ before time α^*_j (Gap A closed), or $i^*(j)$ itself opens by $3\alpha^*_j$ without j''s contribution (condition (D1) met).
- *Spare capacity guarantee.* Show that when j' connects to i', i' has at least one spare slot for j (Gap B closed). A sufficient condition: the LP fractional assignment $x^*_{i'j} > 0$ guarantees j can access i' within the LP's capacity allocation.
- *Factor-revealing LP (for small k).* Encode the two-facility instance (one primary, one alternative) as an LP: $\max T_j/\alpha^*_j$ subject to CS1–CS5, opening conditions for both facilities, and the capacity constraint on i'. Solve for $k = 3, 4, 5$.

OP2. (C2): Dual concentration. (C2) ($\sum_{X^B} \alpha^*_j \le \mathrm{LP}^{y<1}_{\mathrm{slack}}$) holds under (C2a)+(C2b) (Theorem 8.3). Proposition 8.5 shows (C2) is *not* automatic under any algorithm that opens all $y^*_i = 1$ facilities first: the leakage $W(X'_B, F_1) > 0$ in general (Example in the proof of Proposition 8.5).

However, the *weaker* condition (C2′) (Proposition 8.6) suffices for factor 3 under the two-phase design, and is more likely to be provable.

Three approaches to establish (C2) or (C2′):

- *Approach A: Skip Phase 1.5.* $X_{\text{pre}} = \emptyset$: (C2b) holds automatically. (C2a) may still fail, but reduces to $W(X^B, F_{\text{tight}}) = 0$, which holds when F_{tight} spare capacity absorbs all fractional load.
- *Approach B: Two-phase redesign (weakens to (C2′)).* Phase A opens $F_1 = F_{\text{tight}} \cup F^{y=1}_{\text{slack}}$ and assigns $X_1 \cup X^A \cup X_{\text{pre}}$; Phase B is CAPD on X'_B. Factor 3 follows from (C1)+(C2′) (Proposition 8.6). (C2′) requires $\sum_{X'_A} \alpha^*_j \geq (\text{LP} - \text{LP}^{y<1}_{\text{slack}})/3$, i.e., Phase A clients capture at least a third of the $y^*_i = 1$ LP value.
- *Approach C: Enhanced Phase 1.5.* Match into F_{tight} and $F^{y=1}_{\text{slack}}$ spare capacity simultaneously to minimise leakage terms in Theorem 8.3.

OP3. D1 non-primary pairs at $F^{y<1}_{\text{slack}}$ and F_{tight}. Lemma 6.9 resolves D1 for $F^{y=0}$. A unified dual construction across all phases would close the deferred cases.

12 Status Summary

Component	**Claim**	**Status**	**Ref.**
Dual derivation	Correct mechanical dual	Proved	§2
CS1–CS5	Complementary slackness	Proved	§2
Phase 1 capacity	$\lvert X_1(i)\rvert \le (1-\varepsilon)u_i$	Proved	Lem. 4.1
Phase 1 cost	$C^1 \le \mathrm{LP}_{\mathrm{tight}}$	Proved	Lem. 4.2
Phase 1.5 cost	$C^{1.5} \le \mathrm{LP}^{\mathrm{frac}}_{\mathrm{tight}}$	Proved	Thm. 5.2
Joint bound	$C^1 + C^{1.5} \le \mathrm{LP}_{\mathrm{tight}}$	**Proved**	Thm. 5.2
Phases 1+1.5+2a	$\le \mathrm{LP} - \mathrm{LP}^{y<1}_{\mathrm{slack}} \le \mathrm{OPT}$	**Proved**	Thm. 9.1
$\nu_i^* = 0$ for $F^{y=1}_{\mathrm{slack}}$	From CS2	Proved	Lem. 6.1
$S_i \neq \emptyset$ for $F^{y=1}_{\mathrm{slack}}$	Phase 2a opens all $F^{y=1}_{\mathrm{slack}}$	Proved	Lem. 6.3
S_i props.	$\lvert S_i\rvert \le u_i$, disjoint, $c \le \alpha^*$	Proved	Lem. 6.5
Phase 2a cost	$C^{2a} \le \mathrm{LP}^{y=1}_{\mathrm{slack}}$	Proved	Thm. 6.7
Phase 2a dual (primary)	D1–D3 tight	Proved	Thm. 6.7(e)
D1 non-primary ($F^{y=1}_{\mathrm{slack}}$)	$\alpha_j^* \le c(i,j)$	Proved	Lem. 6.8
D1 non-primary ($F^{y=0}$)	LP duals	Proved	Lem. 6.9
D1 non-primary ($F^{y<1}_{\mathrm{slack}}$,F_{tight})	Deferred	Deferred	Lem. 6.9
LP identity	$\sum_j \alpha_j^* x_{ij}^* = \mathrm{LP}(i)$	**Proved**	Lem. 8.1
$\sum_{X^B} \alpha_j^*$	$\le \mathrm{LP}^{y<1}_{\mathrm{slack}} + \text{leakage}$	**Proved**	Thm. 8.3
(C2) under (C2a)+(C2b)	$\sum_{X^B} \alpha_j^* \le \mathrm{LP}^{y<1}_{\mathrm{slack}}$	Conditional	Thm. 8.3
Old TU-only Phase 2b	LP-feasible counterexample	Refuted	Thm. 7.1
Hybrid Phase 2b feas.	All X^B assigned	Proved	Lem. 7.3
Step 2b-i cost	$\le \sum_{X^{B,\mathrm{m}}} \alpha_j^*$	Proved	Lem. 7.4
(C1): CAPD factor 3	$T_j \le 3\alpha_j^*$ for $X^{B,\mathrm{r}}$	**Open**	Conj. 7.5
Factor-3 ratio	$C(S) \le 3\mathrm{OPT}$	Cond. (C1)+(C2)	Thm. 9.2
Factor-3 (weakened)	$C(S) \le 3\mathrm{OPT}$	Cond. (C1)+(C2′)	Thm. 9.4
Factor-4 ratio	$C(S) \le 4\mathrm{OPT}$	Cond. (C1) only	Thm. 9.5
(C2) not automatic	Leakage $W(X'_B, F_1) > 0$ in general	**Proved**	Prop. 8.5
(C2) weakened to (C2′)	Strictly weaker sufficient condition	**Proved**	Prop. 8.6
Proportional align. at $T = 1$	$\tilde{\beta}_{ij}(1) = \beta_{ij}^* + \nu_i^*$	**Proved**	OP**OP1.**
Factor-revealing LP	Encodes worst-case CAPD ratio	Formulated	OP**OP1.**
Old Conj. 7.1	$T_j \le 3\alpha_j^*$ (uniform CAPD)	Retired	§8.7